Junhan "Judy" Kong Curriculum Vitæ

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BIO

Junhan "Judy" Kong is a Ph.D. candidate in the Information School at the University of Washington. She is advised by Prof. Jacob O. Wobbrock and is a member of the DUB Group, CREATE Center, and ACE Lab. She obtained her bachelor's and master's degrees in Computer Science from Carnegie Mellon University with an additional major in Human-Computer Interaction and minors in Machine Learning and Statistics. Her research interests include **human-computer interaction, accessible computing, interaction techniques, and human-centered AI**. Her work approaches ability-based human-AI systems through three pillars:

- *Quantitative modeling of behavior*: establishing quantitative understanding of varying human abilities by modeling multimodal interaction behavior;
- *Enhancing agency in human-AI interaction*: developing effective human-AI collaboration paradigms that enhance user agency and autonomy;
- *Infrastructural support for interface creation*: lowering barriers to creating personalized interfaces through infrastructural support for design and development.

EDUCATION

University of Washington, Seattle, WA — Sep 2020–Jun 2027 (expected)
Ph.D. in Information Science
Thesis (Proposed): Intelligent and Customizable Interfaces for Ability-Aware Accessibility
Advisor: Jacob O. Wobbrock
Dissertation Committee Members: James A. Fogarty, Alexis Hiniker, Shaun K. Kane, Jon E. Froehlich

Carnegie Mellon University, Pittsburgh, PA — May 2019–May 2020
M.S. in Computer Science
Thesis: An Authoring Tool for Creating Interactive AR User Tutorials by Demonstration
Advisor: Jeffrey P. Bigham

Carnegie Mellon University, Pittsburgh, PA — Aug 2015–May 2019
B.S. in Computer Science
Additional major in Human-Computer Interaction, minors in Machine Learning and Statistics

AWARDS AND HONORS

Best Paper Award, MobileHCI 2024
Best Paper Nominee, ASSETS 2022
Special Recognitions for Outstanding Reviews, CHI 2022, UIST 2023 & 2024
Adobe Research Intern Project Expo Winner, Adobe Intern Project Expo 2023
Boeing Blue Skies Award: Game Changer, CMU Undergraduate Research Symposium 2019
University Honors for Academic Excellence, Carnegie Mellon University, 2019
Dean's List, Carnegie Mellon University, School of Computer Science, 2015-2019
Best Educational App, TartanHacks (CMU Hackathon) 2017
Social Impact Prize, TartanHacks (CMU Hackathon) 2016

RESEARCH EXPERIENCES

Research Assistant, University of Washington Sep 2020–current

Advised by Prof. Jacob O. Wobbrock

- Devised new metrics for characterizing fine motor abilities from touchscreen input behavior, conducted experiments to investigate correlations between touch metrics and impairment level as well as specific fine motor challenges [P.2, C.3]. *Received an ACM ASSETS '22 Best Paper Nomination.*
- Created and evaluated a toolkit synthesizing mobile sensor data and providing holistic understandings of user abilities for adaptive mobile interfaces at runtime, simplifying development of ability-aware mobile apps [C.5]. *Received the ACM MobileHCI '24 Best Paper Award.*
- Based on personalized gestures elicited from users with upper-body motor impairments [C.4], developed and evaluated a novel multimodal VR text entry system and proposed a human-AI interaction paradigm of using ability-based AI recommendations to scaffold customization.
- Developed a novel design tool to generate accessibility critique backed by heuristics and personas leveraging LLMs to inform design revisions, and co-created a new design method for conducting accessibility inspections using ability heuristics [C.8].

Research Scientist Intern, Meta Reality Labs Jun 2024–Sep 2024

Mentored by Drs. Amy Karlson, Hemant Surale, Michael J. Proulx

- Devised new measures to quantitatively characterize 3D pointing. Investigated pointing techniques using gaze, hand, and novel ring devices and conducted experiments to validate correlations between new behavioral measures and pointing performance, task load, exertion, and usability [C.9].

Research Scientist Intern, Adobe Jun 2023–Sep 2023

Mentored by Dr. Zoya Bylinskii, Tianyuan Cai

- Investigated the impact of walking on mobile reading, developed a system that provides automatic and customized mobile reading adaptations to walking using smartphone built-in sensors, and investigated the influence of system recommendations on human reading preferences [P.3, C.6].
  Winner of the Adobe Research Intern Project Expo.

Research Assistant, Carnegie Mellon University Aug 2018–Jun 2020

Mentored by Profs. Jeffrey P. Bigham, Anhong Guo, Amy Pavel

- Developed a novel tool for authoring interactive AR tutorials through user demonstration [P.1, C.2].
- Co-designed, developed, and evaluated a system using computer vision to reverse-engineer interfaces from point-of-view videos, improving touchscreen accessibility for visually-impaired users [C.1].
  2nd Place of the CMU Undergraduate Research Symposium, "Boeing Blue Skies Award: Game Changer".

PUBLICATIONS

Peer-Reviewed Conference Papers

[C.9] **Junhan Kong**, Jacqui Fashimpaur, Michael J Proulx, Hemant Bhaskar Surale, Jacob O. Wobbrock, Amy Karlson. Quantifying 3D Pointing: Characterizing What Happens During Pointing Selection in Virtual Reality. In Proceedings of the ACM International Conference on Multimodal Interaction (ICMI '26), October 05–09, 2026, Napoli, Italy. https://doi.org/10.1145/3776574.3831138.

[C.8] Claire L. Mitchell, **Junhan Kong**, Jesse J. Martinez, Shaun K. Kane, Amy J. Ko, Alexis Hiniker, Jacob O. Wobbrock. Ability Heuristics for Conducting Accessibility Inspections. In Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems (CHI '26), April 13–17, 2026, Barcelona, Spain. ACM, New York, NY, USA, 15 pages. https://doi.org/10.1145/3772318.3791504.

[C.7] Xiyuan Shen*, Seokhyun Hwang*, **Junhan Kong**, Alexandre L. S. Filipowicz, Andrew Best, Jean M. Costa, Scott Carter, James Fogarty, Jacob O. Wobbrock. Touchscreens in Motion: Quantifying the Impact of Cognitive Load on Distracted Drivers. In the 38th Annual ACM Symposium on User Interface Software and Technology (UIST '25), September 28-October 1, 2025, Busan, Korea. https://doi.org/10.1145/3746059.3747683.

[C.6] **Junhan Kong**, Jacob O. Wobbrock, Tianyuan Cai, Zoya Bylinskii. Supporting Mobile Reading While Walking with Automatic and Customized Font Size Adaptations. In Proceedings of the ACM SIGCHI Conference on Human Factors in Computing Systems (CHI '25), April 26-May 1, 2025, Yokohama, Japan. https://doi.org/10.1145/3706598.3713367.

[C.5] **Junhan Kong**, Mingyuan Zhong, James Fogarty, Jacob O. Wobbrock. The Ability-Based Design Mobile Toolkit (ABD-MT): Developer Support for Runtime Interface Adaptation Based on Users' Abilities. In Proceedings of the ACM on Human-Computer Interaction (MobileHCI '24). 8, MHCI, Article 277, 27 pages. https://doi.org/10.1145/3676524. ***Best Paper Award***

[C.4] Momona Yamagami, Alexandra A. Portnova-Fahreeva, **Junhan Kong**, Jacob O. Wobbrock, Jennifer Mankoff. How Do People with Limited Movement Personalize Upper-Body Gestures? Considerations for the Design of Personalized and Accessible Gesture Interfaces. In Proceedings of the 25th International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS '23). Association for Computing Machinery, New York, NY, USA, Article 1, 1–15. https://doi.org/10.1145/3597638.3608430.

[C.3] **Junhan Kong**, Mingyuan Zhong, James Fogarty, Jacob O. Wobbrock. Quantifying Touch: New Metrics for Characterizing What Happens *During* a Touch. In The 24th International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS '22), October 23–26, 2022, Athens, Greece. Association for Computing Machinery, New York, NY, USA. https://doi.org/10.1145/3517428.3544804. ***Best Paper Nominee***

[C.2] **Junhan Kong**, Dena Sabha, Jeffrey P. Bigham, Amy Pavel, Anhong Guo. 2021. TutorialLens: Authoring Interactive Augmented Reality Tutorials Through Narration and Demonstration. In Symposium on Spatial User Interaction (SUI '21). Association for Computing Machinery, New York, NY, USA, Article 16, 1–11. https://doi.org/10.1145/3485279.3485289.

[C.1] Anhong Guo, **Junhan Kong**, Michael Rivera, Frank F. Xu, Jeffrey P. Bigham. 2019. StateLens: A Reverse Engineering Solution for Making Existing Dynamic Touchscreens Accessible. In Proceedings of the 32nd Annual ACM Symposium on User Interface Software and Technology (UIST '19). Association for Computing Machinery, New York, NY, USA, 371–385. https://doi.org/10.1145/3332165.3347873.

Posters and Demos

[P.4] Momona Yamagami, Claire L. Mitchell, Alexandra A Portnova-Fahreeva, **Junhan Kong**, Jacob O. Wobbrock. (2024). Personalized gesture classification for encouraging non-sedentary behavior during technology use in people with motor disabilities. Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC '24 Poster). July 15-19, 2024, Orlando, Florida. Piscataway, New Jersey: IEEE Engineering in Medicine and Biology Society.

[P.3] **Junhan Kong**, Tianyuan Cai, Zoya Bylinskii. Improving Mobile Reading Experiences while Walking Through Automatic Adaptations and Prompted Customization. In Adjunct Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology (UIST '23 Poster). Association for Computing Machinery, New York, NY, USA, Article 19, 1–3. https://doi.org/10.1145/3586182.3616666.

[P.2] **Junhan Kong**, Mingyuan Zhong, James Fogarty, Jacob O. Wobbrock. 2021. New Metrics for Understanding Touch by People with and without Limited Fine Motor Function. In The 23rd International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS '21 Poster). Association for Computing Machinery, New York, NY, USA, Article 80, 1–4. https://doi.org/10.1145/3441852.3476559.

[P.1] **Junhan Kong**, Anhong Guo, Jeffrey P. Bigham. 2019. Supporting Older Adults in Using Complex User Interfaces with Augmented Reality. In The 21st International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS '19 Demo). Association for Computing Machinery, New York, NY, USA, 661–663. https://doi.org/10.1145/3308561.3354593.

PATENTS

[IP.1] Anhong Guo, **Junhan Kong**, Michael Rivera, Frank F. Xu, Jeffrey P. Bigham. StateLens: A Reverse Engineering Solution for Making Existing Dynamic Touchscreens Accessible. U.S. Provisional Patent Application 19/207, filed June 6, 2019.

GRANTS AND PROPOSALS CONTRIBUTED

"A design tool for supporting ideation towards improved accessibility through ability heuristic evaluation and persona generation", UW CREATE Seed Funding (proposal submitted; not funded)
PI: Jacob O. Wobbrock 2026
Role: Contributed to grant writing and budget. Proposed project goals, expected contributions, approach and impact, and workplan and timeline.

"Leveraging behavioral signals and generative AI for the rapid creation of interactive, user-adaptive HMI prototypes", Toyota Research Institute ($447,792)
PIs: Jacob O. Wobbrock, James Fogarty, Andrew Best, Alexandre LS Filipowicz, Scott Carter 2024
Role: The proposal built on prior work I led on the development of the Ability-Based Design Mobile Toolkit for adaptive mobile interfaces [C.5]. Contributed to project preparation and launch as the initial student lead.

Gift fund for the ACE Lab, Adobe ($10,000) 2023
Awarded to recognize my Adobe internship work and support follow-up research on mobile reading while walking [C.6] and related conference travel.

INVITED TALKS AND GUEST LECTURES

Ability-Based Human-AI Systems for Personalized Accessibility
University of Michigan, Computer Science and Engineering Oct 2026

Ability-Based Human-AI Systems for Personalized Accessibility
Rochester Institute of Technology, Golisano College of Computing and Information Sciences Oct 2026

Accessibility: What Do Users Really Need?
UW INFO 380 Product and Information Systems Management, Guest Lecture Nov 2025

Readability On-the-Go: Improving Mobile Readability under Situational Impairments
The Readability Consortium Monthly Meeting Jul 2023

Design Thinking and Accessibility
UW IMT 596 MSIM Capstone Project, Guest Lecture Jan 2021

TEACHING

Instructor
Designed and taught the course as the sole instructor, Course Rating: 4.8/5.0
UW INFO 498 Special Topics in Informatics: Accessibility Spring 2024

Teaching Assistant
UW HCID 520 User Interface Software and Technology Winter 2023, 2025, 2026

UW INFO 463 Input and Interaction — Spring 2025
UW INFO 380 Information Systems Analysis and Design — Fall 2020, 2022, 2023, 2025
UW IMT 575 Data Science: Scaling, Applications and Ethics — Spring 2022
UW IMT 596 & 597 MSIM Capstone — Winter 2021, Spring 2021
CMU 05-391 Designing Human-Centered Software — Fall 2017–Fall 2019
CMU 15-122 Principles of Imperative Computation — Fall 2016–Fall 2019

MENTORING

Undergraduate Students

Katelyn Zhao
The Information School, University of Washington — Summer 2026–current

Sanjana Pavani
The Information School, University of Washington — Fall 2023–Fall 2024

SERVICE

Organizing Committee
ASSETS 2025 Student Volunteer Co-Chair
ASSETS 2022 Web and Graphics Design Co-Chair
UW DUB Doctoral Colloquium 2023 Coordinator

Conference Reviewer
UIST (’23, ’24), CHI (’22, ’24, ’25, ’26, ’27), ASSETS (’23 Posters and Demos), SUI (’24), DIS (’24)
Special Recognitions: UIST (’23, ’24), CHI (’22)

Undergraduate Activities
CMU Undergraduate HCI Student Advisory Committee — Sep 2018 - May 2019
CMU Undergraduate Orientation Counselor — Aug 2018

REFERENCES

Jacob O. Wobbrock
Professor, Information School,
University of Washington, Seattle, WA, USA
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James A. Fogarty
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University of Washington, Seattle, WA, USA
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Shaun K. Kane
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Senior Staff Research Scientist, Google Research, Boulder, CO, USA
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Michael J. Proulx
Professor of Cognition and Technology, University of Bath, United Kingdom
Research Scientist, Meta Reality Labs Research, Redmond, WA, USA
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Zoya Bylinskii
Previous Senior Research Scientist, Adobe Research
Founder of Perceptual Insights Inc., Boston, MA
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